

TORSIONVERSE SERIES 2: COUNTER-ROTATING FLUID BELLS—CORIOLIS ROLL AND GYROSCOPIC T_{2g} THRUST

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Companion script: analysis/demos/fluid_bells_doc.py -- 16/16 PASS [AG1-AG3, MC1-MC10]

ABSTRACT

A hand-held or vehicle-mounted thruster using two counter-rotating bell-shaped shells of high-density liquid can, in principle, produce net thrust while imparting zero reaction torque to the operator. The mechanism rests on I_h character-table algebra: the symmetric spin mode (T_{1g}, angular momentum) cancels exactly between counter-rotating units, while the asymmetric mode (T_{2g}, generated by the curved bell wall breaking spherical symmetry) does NOT cancel — because it depends on χ^2 , which is insensitive to the sign of Ω . This is not a new force; it is standard fluid dynamics (Taylor-Proudman boundary layer torque, Coriolis over-roll, hydraulic wedge) amplified through a torsionverse-derived T_{2g}/T_{1g} coupling ratio, plus a genuinely open question of how efficiently that fluid pressure couples to the torsion medium itself (η , not yet derived).

The choice of working fluid matters: torsionverse independently derives (nuclear_geometry.py, not from any external claim) that $Z=114/N=184$ is a doubly-magic nuclear shell closure, and predicts the once-synthesized Mc-299 ($Z=115$, $N=184$) would couple to the T_{2g} thrust mode where mercury's nuclear structure gives exactly zero coupling — a qualitative, not just quantitative, difference.

SECTION 1: COUNTER-ROTATING DEVICE — MECHANISM

Reference: analysis/misc/new_ground.py III.1-III.5, Companion check AG1

Two co-axial bell-shaped or hemispherical shells, each containing a high-density spinning fluid, counter-rotating (Shell 1 at $+\Omega$, Shell 2 at $-\Omega$, both thrust axes aligned).

WHY T_{1g} CANCELS (no torque felt by the operator):

Shell 1: T_{1g} winding, chirality +1 $\rightarrow$ angular momentum $L = +I\Omega$
Shell 2: T_{1g} winding, chirality -1 $\rightarrow$ angular momentum $L = -I\Omega$
Sum: $\Delta L = 0 \rightarrow$ no net torque reaction on the device body / operator hand.
 $\chi(T_{1g_1} + T_{1g_2})$ at C5 = $\phi - \phi = 0$ (exact cancellation) [AG1]

WHY T_{2g} ADDS (net thrust, does not cancel):

Thrust scales as $F \sim \chi^2(T_{2g}) \times \rho_{\text{fluid}} \times (\Omega R)^2$

$\chi^2(T_{2g})$ depends on $|\chi|^2$, not the sign of χ – reversing spin direction does not change the sign of χ^2 . For both units (same geometry, opposite spin):
 $|\chi(T_{2g}, C5)| = 1/\phi$ for BOTH shells
 Net T_{2g} force = $2 \times \chi^2(T_{2g}) = 2/\phi^2 = 0.7639 \rightarrow \text{DOUBLES}$ [AG1]

WHY THE CURVED PATH IS ESSENTIAL:

A symmetric sphere spinning at $\pm\Omega$ is pure T_{1g} – no T_{2g} component, zero thrust even if driven asymmetrically. A bell/paraboloid shape breaks spherical symmetry ($0_h \rightarrow C_{4v}$ or $C_{\infty v}$ via the curved wall's centrifugal-spiral path), which mixes in the T_{2g} component that survives reversal.

THE THREE FLUID-MECHANICS FORCE MULTIPLIERS (standard fluid dynamics, amplifying the torsionverse T_{2g} coupling, from new_ground.py III.1):

1. Boundary layer torque: $T \sim \rho\Omega^2 R^4 f(\text{Re})$ – Taylor-Proudman scaling
 2. Coriolis over-roll: secondary flow amplifies the T_{2g} pressure gradient
 3. Hydraulic wedge: bell curvature reinforces the asymmetric pressure profile
- All three scale as $\Omega^2 \rightarrow$ quadratic RPM curve is the falsifiable signature (new_ground.py III.2): a linear or cubic result would falsify the mechanism.

CONNECTION TO HISTORICAL DEVICES (noted, not claimed as evidence):

Die Glocke descriptions: "two counter-rotating bell-shaped housings containing a rotating mercury-like fluid" match this geometry. This device configuration was derived independently from the torsionverse mechanics; the geometric convergence with historical descriptions is a coincidence worth noting, not supporting evidence – the prediction stands on the mechanics alone.

MEDIUM COUPLING EFFICIENCY η (GENUINELY OPEN):

$\eta = \alpha \times \chi^2(T_{2g}) \times (\text{coupling function of Zone 2 / device scale})$
 This coupling function is NOT yet derived.
 AT LAB SCALE: medium coupling is negligible (medium Bernoulli $\ll$ fluid Bernoulli by ~ 27 orders of magnitude, per gyroscope_doc.py GY2). The mechanism at lab scale is ordinary fluid dynamics amplified by the three multipliers above – not a medium-scale effect.
 AT HIGH-DENSITY / HIGH-RPM REGIMES: Zone 2 coupling may become relevant when the fluid's winding pressure approaches $G_{\text{shear}} \times \gamma_{\text{crit}}$, which requires v_{fluid} approaching $v_s = R \times c = 5.33e7$ m/s – not achievable with any known mechanical system.

ENGINEERING ESTIMATE (fluid dynamics only, no medium coupling assumed):

Liquid mercury, $R=5\text{cm}$, 10,000 RPM: $v_{\text{rim}}=52.4$ m/s, $\Delta P_{\text{fluid}}=1.86e7$ Pa
 F_{wall} per shell ($\chi^2=1/\phi^2$): 55.7 kN (single bell structural load)
 F_{wall} both shells combined ($\chi^2=2/\phi^2$): 111.3 kN (net T_{2g} , both bells)
 The factor of 2 comes from $\chi^2(T_{2g})$ being sign-independent: both shells contribute $+1/\phi^2$ regardless of rotation direction.
 Force scales as Ω^2 (quadratic RPM curve, not linear or cubic – AG2b). These are forces the fluid exerts on the containing structure – a structural engineering number, NOT thrust on external space. The thrust itself requires η (open above).
 Reference: analysis/demos/fluid_bells_doc.py [AG1, AG2a-b, AG3a-b]

SECTION 2: WORKING FLUID — Mc-299 vs MERCURY

Reference: analysis/demos/fluid_bells_doc.py [MC1-MC9, MC10]

Independent derivation (nuclear_geometry.py) — not sourced from any prior popular claim about a specific element.

NUCLEAR SHELL CLOSURE PREDICTION:

Torsionverse independently derives two I_h shell closures from orbital branching rules:

$Z=114$: proton shell closes ($l=6 \rightarrow G_g$ branching, $\text{dim}=14$ basis, NG12)
 $N=184$: neutron shell closes ($l=7 \rightarrow 2 \times (G_g + G_g)$, $\text{dim}=16$ basis, NG13)
 Fl-298 ($Z=114$, $N=184$) is the PURE doubly-magic nucleus (both shells closed, zero unpaired nucleons beyond either closure) – explicitly the "island of stability" prediction in doc_nucleus.txt.
 Mc-299 ($Z=115$, $N=184$) has one proton beyond the $Z=114$ closure – analogous

to Bi-209 sitting one proton beyond the real-world Pb-208 doubly-magic core (Bi-209 alpha half-life ~2e19 years, effectively stable). This is why an "extra" nucleon beyond closure does not necessarily destabilize a nucleus; it depends on which shell (proton vs neutron) the extra nucleon sits in and how far the next open orbital is in energy.

Quantitative analogy (Pb-208/Pb-209/Bi-209, MC10):

Bi-209 (+1 proton beyond Pb-208): $t_{1/2} \sim 2e19$ years

Pb-209 (+1 neutron beyond Pb-208): $t_{1/2} = 3.25$ hours

Ratio: $5.4e22$ — proton-beyond-closure enormously more stable.

Applied here: Mc-299 (proton analog of Bi-209) predicted long-lived;

Mc-300 (N=185, neutron analog of Pb-209) predicted short-lived.

Reference: analysis/demos/fluid_bells_doc.py [MC10a-b]

SHAPE PREDICTION (oblate, not spherical) — MC8:

I_h geometry: the Z=114 (l=6) and N=184 (l=7) preferred axes are offset by the EXACT icosahedral adjacent-C5-axis angle $\arccos(1/\sqrt{5}) = 63.43^\circ$. Stacked proton and neutron orbital planes crossing at this angle create a residual OBLATE deformation that the N=184 shell closure does not cancel (it only cancels the LAST shell's own deformation, not the accumulated crossing of all shells below it).

Predicted: $\beta_{\text{Mc-299}} \approx -0.069$ to -0.112 (oblate, flat-disk shape) — NOT spherical as the naive doubly-magic picture would suggest.

[Falsifiable: a measured Mc-299 quadrupole moment near zero would falsify this axis-crossing correction.]

PHASE STATE PREDICTION (liquid at room temperature) — MC9:

Two bonding scenarios depending on whether the $7p_{1/2}^2$ spin-orbit-split pair acts as an inert electron pair (Hg-like, like $6s^2$ in mercury) or whether all three 7p electrons participate in bonding (Bi-like):

Scenario A (all 7p bond, Bi-like): $T_{\text{melt}} \approx 507 \text{ K} = 234^\circ\text{C} \rightarrow \text{SOLID at RT}$

Scenario B ($7p_{1/2}^2$ inert, Hg-like): $T_{\text{melt}} \approx 211 \text{ K} = -62^\circ\text{C} \rightarrow \text{LIQUID at RT}$

Torsionverse favors Scenario B: the $7p_{1/2}$ - $7p_{3/2}$ spin-orbit splitting at Z=115 (~3.5-4 eV) is ~140x larger than kT at room temperature, strongly favoring an inert pair — the same mechanism that makes mercury ($6s^2$ inert pair) liquid at RT. Predicted: Mc-299 is a liquid metal at RT, melting near -62°C , density ~13,500 kg/m³ (mercury-comparable).

[This directly disagrees with standard relativistic DFT literature, which predicts ~400 K / solid at RT — a genuine falsifiable difference.]

PROTOCOL BETA COUPLING COMPARISON:

Mc-299's unpaired 115th proton occupies the $2f_{7/2}$ orbital, predominantly G_g character ($\chi(\text{G}_g, \text{C5}) = -1$). Mercury's last proton (Z=80) occupies $1h_{11/2}$, predominantly H_g character ($\chi(\text{H}_g, \text{C5}) = 0$ — the 5-fold symmetric ring sums to zero at C5).

$\chi(\text{G}_g \times \text{T}_{2g}, \text{C5}) = (-1) \times (-1/\phi) = +1/\phi = +0.618$ (CONSTRUCTIVE)

$\chi(\text{H}_g \times \text{T}_{2g}, \text{C5}) = 0 \times (-1/\phi) = 0$ (ZERO coupling)

This is a QUALITATIVE difference, not merely quantitative: mercury's nuclear structure gives exactly zero coupling to the T_{2g} thrust mode by I_h symmetry; Mc-299 gives a nonzero, constructive coupling. [MC5-MC7]

The oblate nuclear shape (MC8) also geometrically aligns with the rotating bell's own T_{2g} symmetry — a flat-disk nucleus in a device built around a T_{2g}-symmetric flow pattern, vs. mercury's spherical nucleus having no preferred axis to align with.

LOW-DENSITY FLUID / MICROGRAVITY NOTE:

High-density fluids (mercury, Mc-299) are needed on Earth to overcome gravity. In a frictionless environment (orbit, deep space) the threshold drops to zero: $F_{\text{net}} = F_{\text{T2g}} - F_{\text{friction}}$, and friction is absent. Even water ($\rho \approx 1,000 \text{ kg/m}^3$, roughly 1/13.5 of mercury) produces a nonzero T_{2g} pressure gradient by the same χ^2 algebra — the scaling is geometric, not a property of the fluid. The force is smaller by $\sim \rho_{\text{water}}/\rho_{\text{Hg}} \approx 7.4\%$, but in microgravity any nonzero net force accumulates without limit over time. A hand-held water-filled counter-rotating device at low RPM could let an astronaut set a travel vector without expending propellant. The χ^2 coupling and $F \propto \Omega^2$ scaling (AG2b) hold for any Newtonian fluid.

THRUST MECHANISM NOTE (handheld case):

The handheld water bell works via STRUCTURAL FORCE ($F_{\text{wall}} \times \eta$), not field

envelopment. The T_{2g} depletion zone from a 5cm bell extends only ~5-10 cm – the astronaut's body is NOT inside it. The bell exerts force on its own wall structure; this is transmitted mechanically to the astronaut's hand. The result is a FELT thrust (attitude control), not free fall. This is the correct and useful behavior for attitude control: the astronaut points the device and feels the push, like a thruster with no propellant. Full Protocol Beta free-fall behavior requires the craft to be enclosed by or adjacent to bells large enough to envelop it. See docs/series2/doc_series2_spinning_pyramid.txt Section 8.6 for the field-size and envelopment analysis.

Quantitative estimate (R=5cm bell, water): [AG3, 5/5 PASS]
 $F_{\text{water}}/F_{\text{Hg}} = \rho_{\text{water}}/\rho_{\text{Hg}} = 7.39\%$ exactly (chi algebra is fluid-agnostic)
 F_{wall} at 100 RPM (wrist speed): 0.82 N
 $F_{\text{wall}} = 1$ N benchmark: crossed at 110 RPM – easily hand-operated
 Illustrative ($\eta=0.01$, 80 kg suited astronaut, 1 h at 100 RPM):
 $a \approx 0.10 \text{ mm/s}^2$, cumulative $\Delta v \approx 0.37 \text{ m/s}$
 Note: η is genuinely open (GY2); these are order-of-magnitude estimates.
 Reference: analysis/demos/fluid_bells_doc.py [AG3a-b]

SECTION 3: OPEN ITEMS

GENUINELY OPEN:

- Medium coupling efficiency η (fluid pressure → torsion medium transfer): the central unresolved question for whether Protocol Beta produces any thrust on external space at all, vs. purely internal structural forces.
- Mc-299 synthesis path to the doubly-magic isotope (current experiments reach Mc-287 to Mc-291, roughly 8+ neutrons short of N=184).
- Mc-299 vs Mc-300 (N=185) stability: Pb/Bi analogy quantified [MC10, CLOSED].
- T_{2g} asymmetric rotor thrust – quantitative first-principles derivation beyond the χ^2 -scaling argument (an actual force-per-unit-area formula).
- NULL ZONE ARRAY GEOMETRY FOR THRUST (shape TBD):
 An array of orthogonal T_{1g} beam pairs creates a jammed cell shell surrounding a depleted interior. If the shell geometry is asymmetric (e.g., a hemisphere above the craft with open bottom), the pressure differential between the locked shell and the depleted interior would produce a net upward force.

PULSED OPERATION (KEY INSIGHT): permanent depletion is NOT required. Short high-intensity pulses keep cells in the elastic regime -- they snap back after each pulse (on the cell oscillation timescale, ~30 ys, not $\tau_{\text{relax}} = 3.2$ years). Protocol for continuous lift:

1. Pulse the orthogonal-beam array (elastic, ns-scale pulse)
2. During pulse: A_g locking creates jammed shell + high-intensity interior. Maximum pressure differential → net upward force on craft.
3. Pulse off: cells snap back (elastic, near-instantaneous)
4. Repeat immediately at high repetition rate (kHz to GHz)

Time-averaged force = $F_{\text{pulse}} \times \text{duty_cycle}$ = constant lift.
 Duty cycle controls throttle. No exotic materials; no sustained depletion.

BENCH TEST: see experiments/bench_test_orthogonal_waves.txt
 OPEN: asymmetric shell geometry for unidirectional thrust, force calculation.

ESSENTIALLY CLOSED (pending companion script):

- [x] T_{1g} cancels, T_{2g} doubles for counter-rotating pair – exact chi algebra [AG1]
- [x] Curved bell wall required to generate T_{2g} (symmetric sphere gives none)
- [x] Mc-299 G_g orbital constructively couples to T_{2g} thrust; mercury's H_g orbital gives exactly zero coupling [MC5-MC7]
- [x] Mc-299 predicted oblate ($\beta \approx -0.07$ to -0.11), not spherical [MC8]
- [x] Mc-299 predicted liquid at RT (~-62°C melting), mercury-comparable density – a falsifiable disagreement with standard DFT [MC9]

COMPANION SCRIPT

analysis/demos/fluid_bells_doc.py -- 16/16 PASS [AG1-AG3, MC1-MC10]

PRIOR ART

Closest existing EVA attitude control devices:

SAFER (Simplified Aid For EVA Rescue, NASA, 1994-present):

24 cold-gas nitrogen thrusters in a backpack. Expendable propellant; single-use emergency device. Produces thrust by expelling mass.

MMU (Manned Maneuvering Unit, NASA, 1984, retired):

Larger cold-gas unit, same expulsion principle.

Reaction wheels / Control Moment Gyroscopes (CMGs):

Used on spacecraft (ISS, Hubble) for attitude control. Solid rotating masses. Not used on suited astronauts. No fluid component.

Fluid-ring gyroscopes:

Exist for sensing (inertial navigation), not for propulsion.

No publicly documented counter-rotating sealed-fluid bell device for astronaut attitude control or propulsion is known in the literature.

What distinguishes the present mechanism (if $\eta > 0$ is confirmed):

- No propellant consumed: fluid is sealed and recirculating
- No exhaust plume: safe for proximity operations near crew/structures
- Zero net torque on operator: T_{1g} cancels exactly (AG1), unlike a single-rotor gyroscope which torques the user's wrist
- Throttleable by RPM: force $\propto \Omega^2$ (AG2b), continuous analog control
- Buildable with off-the-shelf parts at the mechanical level regardless of whether the torsion-medium thrust (η) is confirmed

The "Die Glocke" historical claim (counter-rotating drums of heavy liquid, alleged 1940s Germany) is the only prior narrative overlap. That claim has no verified engineering documentation and no confirmed operational result. The present paper provides the first chi-algebra mechanical explanation for why such a geometry would cancel operator torque while retaining T_{2g} force.

EXTERNAL INPUTS:

Mercury density/melting point [standard reference values]

Bismuth density/melting point, spin-orbit splitting estimate [literature]

Relativistic DFT Mc melting point comparison [literature, for contrast]

TORSIONVERSE-DERIVED:

$\chi(T_{1g}, C5) = \phi$, $\chi(T_{2g}, C5) = -1/\phi$ [I_h character table]

$Z=114$, $N=184$ shell closures [nuclear_geometry.py, independent of any external claim about a specific element]

I_h adjacent $C5$ axis angle = $\arccos(1/\sqrt{5}) = 63.4349^\circ$ [exact icosahedral geometry]